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Precomputed Azimuth–Elevation Horizon Masks for Satellite Visibility over Terrain

A Spatial-Coherence-Exploiting Lookup Structure

LATEX → PDF · ~23 PAGES, 557 KB

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§ Abstract

We present a complete methodology for constructing a lookup table that maps a geodetic query point (λ, ϕ) to the terrain-induced azimuth–elevation mask governing satellite visibility from that location. The mask $\mathcal{H}_{\lambda\phi} : S^1 \to [-\frac{\pi}{2}, \frac{\pi}{2}]$ encodes, for every azimuth φ, the minimum elevation above which the sky is unobstructed by terrain. A satellite at topocentric direction (φ^*, η^*) is visible iff $\eta^* > \mathcal{H}_{\lambda\phi}(\varphi^*)$.

Naïve per-query computation on a DEM of n cells costs $\Theta(n^{1.5})$ per point, $\Theta(n^{2.5})$ for the all-cells horizon. We review the classical R2, R3, and XDraw viewshed algorithms, Stewart's $O(n \log n)$ all-points horizon algorithm, and the sDEM / band-of-sight data relocation technique of Tabik et al., then build a lookup structure that exploits the strong spatial coherence of horizons between nearby observers.

The central observation is a *far/near decomposition*: the horizon at a point $\boldsymbol{p}$ is the pointwise maximum of a *far* component, governed by distant terrain and nearly constant over tiles of diameter $\ll R_{\max}$, and a *near* component, which varies rapidly but can be computed on demand from a small local window. Combined with low-rank (PCA) per-tile residuals and piecewise-linear azimuthal encoding, total storage drops 1–2 orders of magnitude relative to naïve per-cell storage, with bounded error characterised in closed form.

CONTENTS

§ Problem statement

Given a digital elevation model $\mathcal{T} : \mathcal{D} \to \mathbb{R}$ on a bounded geographic domain $\mathcal{D}$ with cell size Δ and a maximum sensing range $R_{\max}$, construct a data structure $\mathcal{M}$ supporting queries

$$\mathcal{M}(\lambda, \phi) \longmapsto \widehat{\mathcal{H}}_{\lambda\phi}(\cdot), \qquad \widehat{\mathcal{H}}_{\lambda\phi} : S^1 \to \mathbb{R},$$

such that $\|\widehat{\mathcal{H}}_{\lambda\phi} - \mathcal{H}_{\lambda\phi}\|_\infty \leq \varepsilon_{\mathrm{tol}}$ for every $(\lambda, \phi) \in \mathcal{D}$, using as little storage and query time as possible.

§ Contributions

- A uniform mathematical formulation admitting curvature and refraction in closed form, relating the az-el mask to the classical viewshed.
- A consolidated exposition of R2, R3, XDraw, Stewart's sweep, and Tabik's sDEM with precise complexities on a common grid model.
- A quantitative analysis of spatial coherence yielding a rigorous far/near decomposition theorem (Theorem 5.1) with explicit Lipschitz bound $|\Delta\mathcal{H}^{\text{far}}| \leq \delta(1+M)/R_n + \text{curvature term}$.
- A lookup structure built from hierarchical tiles carrying compressed far-horizons plus lazy near-horizon evaluation, with provable error bounds.
- End-to-end specification for converting orbital state to az-el and querying the mask, including refraction and curvature corrections appropriate for VHF–Ka radio and optical sensing.

§ The gap this fills

The viewshed literature divides into (i) single-observer viewshed algorithms (R2/R3 of Franklin & Ray, XDraw of Wang et al.), (ii) all-points total-viewshed / horizon-map algorithms (Stewart, De Floriani & Magillo, Tabik et al., Sanchez-Fernandez et al.), and (iii) GPU / I/O-efficient implementations (Cauchi-Saunders & Lewis, Fishman et al., Qarah). The satellite-specific specialisation appears in mission-analysis tools — AGI STK's `AzElMask` / `TerrainMask` constraints, MATLAB's Satellite Communications Toolbox visibility-mask workflow, ground-station design literature. Solar-resource and shading work is structurally the same problem. *None of these, to the author's knowledge, addresses the offline construction of a compressed $(\lambda, \phi) \mapsto \mathcal{H}$ lookup using formal far/near decomposition with closed-form error bounds.* That is the gap.

§ Status & provenance

Produced in conversation 19824ab5 on 2026-04-23 alongside the broader viewshed-compute Python package. Compiled cleanly (0 errors) after working through several macro-conflict issues during the run — accidental backspace bytes from `\b*` macros and a few `\Rearth` / `\Return` collisions.